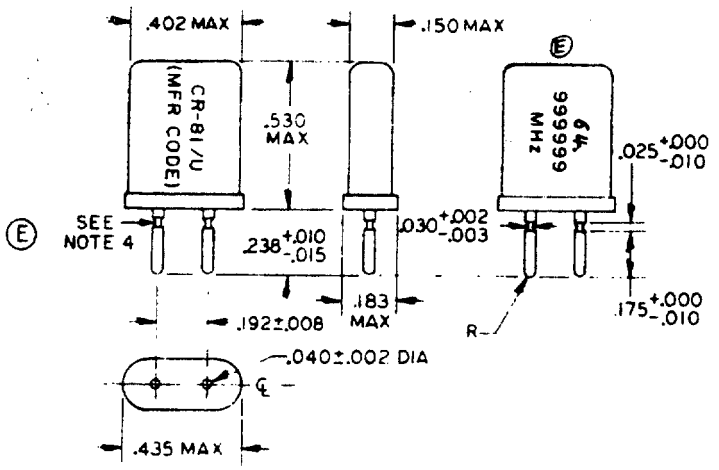
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-81/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 17 to 65 MHz; third overtone; noncontrolled; series.



INCHES	MM	INCHES	MM
.002	.05	.150	3.81
.003	.08	.175	4.45
.008	.20	.183	4.65
.010	.25	.192	4.88
.015	.38	.238	6.05
.025	.64	.402	10.21
.030	.76	.435	11.05
.040	1.02	.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designation, frequency, and code designating characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.
(E) 4. The pin undercut may be omitted.

(E) denotes change

FSC 5955

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TABLE I. Requirements.

Requirements	Specified value or condition
Holder	HC-25/U (MIL-H-10056/11)
Dimensions, marking, and configuration	See figure
Frequency range	17 to 65 MHz, inclusive
Capacitance, shunt	7 pF, maximum
Frequency tolerance (operating temperature range)	±50 PPM
Equivalent resistance	40 ohms, maximum
Mode of oscillation	Third mechanical overtone
Government reference standard test set	TS-683/TSM
Calibration values:	
Resistance	40 ohms
Resistor voltage drop	0.28 volt
Drive level, rated	2.0 ± 0.4 mW
Temperature range:	
Operating	-55° to +105°C, inclusive
Shock (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent-resistance change permitted	±10 percent
Vibration, high frequency 1/:	
Frequency change permitted	±5 PPM
Equivalent-resistance change permitted	±10 percent
Temperature run:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	5 PPM

1/ Method 204 of MIL-STD-202, test condition A.

NOTE: NATO preferred type NXT43.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-0408-31)

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